Muhammad Moiz Bukhari

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EXPERIENCE

Systems Engineering CO-OP Jan 2025 – Aug 2025
Abbott Laboratories *Pleasanton, CA*

- Designed and built a 5-fixture Power Automation system from a blank page, combining embedded Arduino C/C++ firmware with Python test sequencing to run 72-hour unattended endurance cycles with zero manual intervention.
- Architected modular firmware control logic that decoupled sensor drivers, relay/actuator control, and communication layers across 5 fixture revisions, establishing a reusable pattern that cut rework on every subsequent build.
- Engineered embedded control systems in Arduino C/C++ — relay boards, color sensors, LCD status displays, and custom battery-detection algorithms — taking automated power-cycling from concept to production-ready.
- Executed fixture hardware builds single-handedly from design to production: soldering, wiring, crimping, and continuity/voltage validation against schematics, delivering builds ready for Long-Term Testing without rework cycles.
- Drove FDA-compliant V&V execution on the HeartMate Touch Monitor, independently running formal test protocols across multiple BLE pairing scenarios to verify system-requirement compliance.
- Authored IQ/OQ validation packages and CO/CA records in Windchill PLM, establishing the documentation standard for the fixture program and ensuring full audit traceability under FDA 21 CFR 820 and ISO 13485.
- Built Python-based test infrastructure with real-time logging and anomaly detection across 4+ parallel fixture configurations, serving as the go-to engineer for cross-team debug and root-cause resolution.

Software Engineer May 2026 – Present
TWM Wholesale *Stockton, CA*

- Design and build a full e-commerce platform from scratch, including user authentication and server-side data encryption, supporting a catalog of 40,000+ SKUs.
- Eliminate 3-4 hours/day of manual data entry by designing and deploying 10+ Python automation scripts across inventory and procurement workflows.
- Improve automation uptime by tracing recurring failure points through log analysis and rebuilding scripts as fault-tolerant pipelines.

PROJECTS

WaterCon — Cloud-Connected IoT Platform
Personal Project

- Designed and built RESTful APIs and a normalized SQL schema for real-time telemetry across 3+ sensor device types, reducing stakeholder reporting time from hours to seconds.
- Delivered 99.5% platform uptime through fault-tolerant Python ingestion pipelines with automated anomaly detection and alerting, built solo end-to-end.

E2EE Messenger — Team Project

- Collaborated on a full-stack, privacy-first encrypted messaging app using React Native (Expo) and Go, implementing the Signal protocol (X3DH key agreement and Double Ratchet) for end-to-end encryption.
- Built real-time WebSocket messaging with auto-reconnect and safety-number key verification for contact identity confirmation.

SKILLS

Core Languages: Python, C/C++, C#, SQL, Bash, PowerShell, MATLAB
Embedded & Systems: Arduino/AVR, UART/SPI/I2C, GPIO, ADC/PWM, Relay Control, BLE, Firmware Architecture
Backend & Data: RESTful API Design, Relational DB Design, Query Optimization, ETL Pipelines
Regulated & Quality: FDA 21 CFR 820, ISO 13485, IEC 62304, V&V Protocols, IQ/OQ/PQ Validation
Platforms & Tools: Linux (Ubuntu, CentOS), Git/BitBucket, VMware, Windchill PLM
Hardware & Lab: Soldering, Wire Stripping/Crimping, Continuity/Voltage Testing, Schematic Reading, SolidWorks
Process: Agile/Scrum, Certified Scrum Master, Cross-Functional Collaboration, Root-Cause Analysis, Technical Documentation
Additional Languages: Java, JavaScript, PHP

EDUCATION

University of the Pacific Stockton, CA
B.S. Computer Science, Honors. Dean's List (All Semesters). GPA: 3.83. *Dec 2025*